Crank-slider mechanism

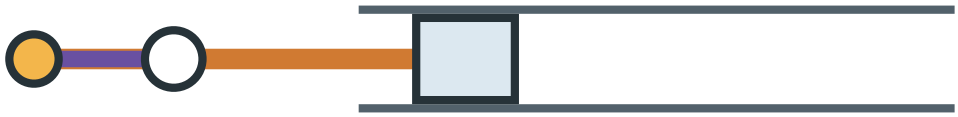
This schematic shows a crank pin following one full circular path and driving a piston through a fixed-length connecting rod. Its time parameterization is non-uniform; the equations use only existing AMX arithmetic, abs, conditional expressions, and sqrt, with no trigonometric functions.

```
let crankRadius: Number = 34
let rodLength: Number = 105
let pivotX: Number = 70
fn crankX(t: Number, r: Number): Number = r * (1 - 2 * abs(t - 1))
fn crankY(t: Number, r: Number): Number = if t <= 1 then sqrt(r * r - crankX(t, r) * crankX(t, r)) else -sqrt(r * r - crankX(t, r) * crankX(t, r))
fn sliderX(t: Number, pivot: Number, r: Number, rod: Number): Number = pivot + crankX(t, r) + sqrt(rod * rod - crankY(t, r) * crankY(t, r))
diagram crankSlider {
title: "Crank-slider mechanism"
description: "A crank pin follows one full circular path and drives a piston through a fixed-length connecting rod."
width: 320
height: 200
time t from 0 to 2 duration 4
line lowerGuide {
x1: 115
y1: 88
x2: 260
y2: 88
stroke: "#52616B"
lineWidth: 2
arrow: "none"
}
line upperGuide {
x1: 115
y1: 112
x2: 260
y2: 112
stroke: "#52616B"
lineWidth: 2
arrow: "none"
}
line connectingRod {
x1: pivotX + crankX(t, crankRadius)
y1: 100 + crankY(t, crankRadius)
x2: sliderX(t, pivotX, crankRadius, rodLength)
y2: 100
stroke: "#D07A32"
lineWidth: 5
arrow: "none"
}
line crank {
x1: pivotX
y1: 100
x2: pivotX + crankX(t, crankRadius)
y2: 100 + crankY(t, crankRadius)
stroke: "#6950A1"
lineWidth: 4
arrow: "none"
}
circle pivot {
cx: pivotX
cy: 100
r: 7
stroke: "#263238"
fill: "#FFFFFF"
lineWidth: 2
}
circle crankPin {
cx: pivotX + crankX(t, crankRadius)
cy: 100 + crankY(t, crankRadius)
r: 6
stroke: "#263238"
fill: "#F3B64B"
lineWidth: 2
}
rectangle piston {
x: sliderX(t, pivotX, crankRadius, rodLength) - 12
y: 90
width: 24
height: 20
```

```
stroke: "#263238"
fill: "#DCE8F0"
lineWidth: 2
}
text timeLabel {
x: 18
y: 24
value: "t = ${round(t, 2)} s"
fill: "#263238"
}
}
```

Crank-slider mechanism

A crank pin follows one full circular path and drives a piston through a fixed-length connecting rod.



t = 0 s

The crank equations parameterize a complete revolution over `t = 0` to `t = 2`. The rod-length constraint determines the slider position directly for every selected time; the document does not model accumulated simulation state.